

all major 3GPP frequency bands. Keysight's 5G Protocol Conformance Toolset offers a number of Global Certification Forum (GCF)-validated 5G NR protocol test cases in non-standalone (NSA) mode across both frequency ranges of sub-6GHz (FR1) and mmWave (FR2). It also offers protocol test cases for standalone (SA) mode as validated by both GCF and PCS-Type Certification Review Board (PTCRB).

Keysight's E5080B, P50xxA series and M980xA series of network analysers deliver the core functionality needed for engineers to perform the required measurements.

mmWave vector signal transceiver (VST). It addresses the test challenges of 5G mmWave RFIC transceivers and power amplifiers. It combines RF and baseband vector signal analyser and generator with a user-programmable FPGA for real-time signal processing in 5G's mmWave. National Instruments' (NI's) PXIe-5831 VST addresses time-to-market challenges for X-band, Ku-band and Ka-band radar and satellite communications components and systems.

NI's Semiconductor Test System (STS) includes a multi-site mmWave test STS configuration optimised for 5G power amplifiers, beamformers and transceivers. It lets users perform faster and optimised measurements, inline signal processing and high-speed data transfers.

Analog-to-digital converter (ADC). Texas Instruments (TI) has introduced an ultra-high-speed ADC (12DJ5200RF) with 8GHz bandwidth, 10.4GSPS sampling rate and low power consumption. This helps achieve high measurement accuracy for 5G testing applications and oscilloscopes, and direct X-band sampling for radar applications.

Radio communication test station. An anechoic chamber is a validated test method to conduct 5G measurements. Optimised implementation of Anritsu's MT8000A with CATR chamber for beam characterisation and verification significantly reduces overall calibration time, cost and complexity of measurement setup with NSA and SA base station simulation functionality for developing 5G chipsets. It also supports all frequency bands needed during initial 5G deployments—2.5GHz, 3.5GHz and 4.5GHz in FR1 band, and 28GHz and 39GHz in FR2 band.

Surge in demand for 5G testers

The number of trials for effective 5G implementation has been on a constant rise. This is a key reason for increase in demand for 5G testers in the global market. Use of GCF-certified 5G chipsets, modems and modules will ensure reliability of service and successful deployment of 5G communications and the IoT services. This will be critical for all applications, from driverless cars, factory and process automation to augmented, virtual and mixed reality (AR, VR and MR). The market is ready for its first set of 5G devices; however, service providers and network equipment manufacturers will need to keep working on their improvement areas. **EFY**

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